

Species-Aware Curation Review:

Prokaryotic Molybdenum Cofactor (Moco) Biosynthesis in *Pseudomonas putida* KT2440

Module scope: GTP → cyclic pyranopterin monophosphate (cPMP) → molybdopterin (MPT) → Mo-MPT, plus the optional dinucleotide variants MGD and MCD. **Target:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556). **Seeding bucket:** KEGG `ppu00790` ("Folate biosynthesis"); KEGG module `ppu_M00880` ("Molybdenum cofactor biosynthesis, GTP => molybdenum cofactor").

1. Executive summary

The generic prokaryotic Moco module is **fully satisfiable** in *P. putida* KT2440. Every catalytic step from GTP to Mo-MPT, and **both** optional dinucleotide branches (MGD via MobA and MCD via MocA), is encoded. Mapping to the authoritative resources (UniProt UP000000556 + KEGG orthology) gives a complete, one-gene-per-step assignment with several curation-relevant refinements:

- **Two steps that look absent in the seeded candidate list are in fact covered:** the sulfur-relay genes **moeB (PP_0735)** and **moaD (PP_1293)** were omitted from the 32-gene candidate set because they are bucketed under KEGG `ppu04122` (sulfur relay), not `ppu00790`.
- **No dedicated MogA exists in KT2440.** The MogA-type MPT-adenylyltransferase step (EC 2.7.7.75) is carried by the **MoaB lineage** (PP_2122, PP_4600) — the "catalytically competent prokaryotic MoaB" alternative already anticipated by the module. This is the only step with residual mechanistic uncertainty (`candidate_uncertain`).
- **Two genes labelled generically as "MobA-like NTP transferase" (PP_2483, PP_4230) are specifically MocA** cytidylyltransferases (K07141, EC 2.7.7.76) — the MCD branch. KT2440 is a well-documented heterologous host that matures **MCD-form** molybdenum hydroxylases, giving direct target-strain support.

- **The seeding bucket is contaminated:** of 32 candidates, only ~11 are Moco genes; the rest are folate, queuosine (7-deazapurine), riboflavin and biopterin genes that share pterin/GTP chemistry but belong to neighbouring pathways.
- KT2440 shows **paralog expansion** (3× MoaA, 2× MoaB, 2× MocA) organized as a housekeeping core plus **molybdoenzyme-adjacent gene islands** that supply client-specific cofactor tailoring.

Bottom line for curation: mark all core steps `covered`; mark the MPT-adenylylation step `covered` with a `candidate_uncertain` mechanistic flag (MoaB-for-MogA); no steps are `gap` or `not_expected_in_target_taxon`; the generic module boundaries are correct but the *ppu00790* seeding bucket is too broad and needs pruning.

2. Target-organism pathway definition

Included process (this module): the biosynthesis of the pyranopterin dithiolene ligand from GTP and its loading with molybdenum, i.e.

1. radical-SAM cyclization of GTP to 3',8-cyclo-7,8-dihydro-GTP (MoaA);
2. rearrangement to cPMP (MoaC);
3. activation of the MoaD sulfur carrier (MoeB) and double sulfur insertion into cPMP to give MPT (MoaD–MoaE synthase);
4. adenylylation of MPT (MogA-type activity) and molybdate insertion to give Mo-MPT (MoeA);
5. optional attachment of GMP → MGD (MobA) or CMP → MCD (MocA).

Explicitly excluded (and must be kept in separate modules): molybdate uptake (**modABC** PP_3828–3830; regulator **modR** PP_0360), upstream sulfur mobilisation, terminal cofactor sulfuration (e.g. **fdhD**-type sulfurtransferase PP_0257 for formate dehydrogenase), cofactor insertion into apo-enzymes (XdhC-family chaperones PP_2480, PP_4231, PP_4280, xdhC PP_4279–4280 region), the mature molybdoenzyme reactions themselves (nitrate reductase PP_1703, formate dehydrogenases fdoG PP_0489/PP_2185, nicotinate dehydrogenase nicB PP_3948, xanthine dehydrogenase PP_4234/PP_4279, isoquinoline/quinoline oxidoreductases PP_2478/PP_3622, MsrP PP_4676), and pathway regulation.

Neighbouring pathways to keep separate (critical here): KEGG map00790 is an umbrella "folate/pterin" map, not folate-specific. It bundles four biochemically distinct GTP/pterin branches that must not be merged into the Moco module: - **Folate/tetrahydrofolate:** folE1/E2

(GTP cyclohydrolase I), folB, folK, folP, folC, folA, folM, pabB, pabC, PP_0393 → map00790/map00670/map04981. - **7-deazapurine / queuosine**: queC, queD, queE, queF → 7-deazaguanine branch of map00790. - **Riboflavin**: ribA, ribAB-I, ribAB-II (GTP cyclohydrolase II) → map00740. - **Biopterin / pterin regeneration**: phhA (Phe hydroxylase), phhB (pterin-4 α -carbinolamine dehydratase) → map00360/aromatic amino-acid metabolism.

Database-specific names / IDs: KEGG module **M00880**; KEGG maps map00790 (Folate biosynthesis), map01240 (Biosynthesis of cofactors), map01100 (Metabolic pathways); MetaCyc "molybdenum cofactor biosynthesis" (PWY-5963/PWY-6823); GO:0006777 (Mo-molybdopterin cofactor biosynthetic process). Common gene-name synonyms: MoaA=MoaA/moco-A; cPMP was historically called "precursor Z"; MPT = molybdopterin/pterin; MGD = MPT-guanine dinucleotide; MCD = MPT-cytosine dinucleotide.

3. Expected step model and satisfiability

KEGG module **M00880** definition: ((K03639 K03637),K20967) (K03635,K21142) (((K03831,K03638) K03750),K15376) — i.e. (MoaA+MoaC) → MoaE synthase → (MogA/MoaB adenylyltransferase → MoeA molybdotransferase). MoaD (K03636), MoeB (K21029) and the dinucleotide transferases (K03752 MobA, K07141 MocA) are captured in the sibling sulfur-relay/dinucleotide steps.

#	Module step	EC / KO	KT2440 gene(s)	Status
1	MoaA GTP 3',8-cyclase (radical-SAM)	4.1.99.22 / K03639	PP_4597 (canonical) + PP_1969, PP_2482 (paralogs)	covered (3 copies)
2	MoaC cPMP synthase	4.6.1.17 / K03637	PP_1292	covered
3	MoeB MoaD-adenylyltransferase (sulfur-carrier activation)	2.7.7.80 / K21029	PP_0735	covered (missing from candidate list)
4	MoaD sulfur carrier	– / K03636	PP_1293	covered (missing from candidate list)
5	MoaD–MoaE molybdopterin synthase	2.8.1.12 / K03635	PP_1294 (+PP_1293)	covered
6	MPT adenylylation (MogA-type)	2.7.7.75 / K03638	PP_2122 (moaB-I), PP_4600 (moaB-II) — no dedicated MogA	covered / candidate_uncertain
7	MoeA molybdotransferase	2.10.1.1 / K03750	PP_2123	covered
8	MobA → MGD (optional)	2.7.7.77 / K03752	PP_3457	covered
9	MocA → MCD (optional)	2.7.7.76 / K07141	PP_2483, PP_4230	covered (branch active by direct evidence; gene→MocA call orthology/context-based, moderate confidence)

Steps missing from metadata but present: #3 MoeB (PP_0735) and #4 MoaD (PP_1293) — both real, both in the moaCDE operon / sulfur-relay bucket. **Steps not expected / not applicable:** none are absent. The eukaryotic MOCS/CNX/gephyrin fusion organisation (K15376, K20967, K21142) is correctly **not_expected_in_target_taxon** (bacteria use discrete genes).

4. Candidate genes and evidence

Genomic architecture (KEGG positions). Two organisational tiers: - **Housekeeping core:** `moaC-moaD-moaE` contiguous operon (PP_1292/1293/1294, overlapping start-stop codons); `moaB-I-moeA` operon (PP_2122/2123, overlapping); standalone `moeB` (PP_0735); canonical `moaA` (PP_4597) beside `moaB-II` (PP_4600) and an oxidoreductase (PP_4596); standalone `mobA` (PP_3457). - **Molybdoenzyme gene islands:** a MoaA paralog + MocA pair (PP_2482 + PP_2483) embedded beside an isoquinoline-oxidoreductase/XDH island (PP_2478, PP_2480); a second MocA (PP_4230) contiguous with xanthine-dehydrogenase genes (PP_4231, PP_4234). This physical linkage supports **client-specific cofactor tailoring** rather than redundancy.

High-confidence genes (direct SwissProt-level or KO evidence): - **PP_4597 moaA** (Q88E69, SwissProt): 4Fe-4S radical-SAM GTP 3',8-cyclase; well-annotated. Caveat: KEGG's *GenBank product name* mislabels it "cyclic pyranopterin monophosphate synthase" — rely on KO/UniProt, not GenBank text. - **PP_1292 moaC** (Q88NC0, SwissProt): cPMP synthase; single copy; unambiguous. - **PP_3457 mobA** (Q88HA3, SwissProt): MGD synthase EC 2.7.7.77; single copy; unambiguous. - **PP_0735 moeB** (Q88PW3): MoaD-adenylyltransferase EC 2.7.7.80 (K21029); high confidence. - **PP_2123 moeA** (Q88L14): molybdotransferase EC 2.10.1.1 (K03750); high confidence. - **PP_1294 moaE / PP_1293 moaD** (Q88NB8 / Q88NB9): MPT-synthase large/small subunits; operon context reinforces.

Paralog-ambiguous / broad-mapping genes (curation attention): - **MoaA ×3 (PP_4597, PP_1969, PP_2482):** all K03639. PP_4597 is the canonical housekeeping copy; PP_1969 and PP_2482 are paralogs (PP_2482 is island-embedded). Broad GO/EC transfer risk: paralogs may be over-annotated as fully interchangeable; functional specialisation is inferred, not proven. - **MoaB ×2 (PP_2122, PP_4600):** both K03638 / EC 2.7.7.75. **Domain evidence confirms they are bona fide MoaB, not MogA:** each carries the MoaB-specific InterPro signatures IPR012245 (MoaB) and IPR013484 ("Molybdenum cofactor biosynthesis protein B, proteobacteria") on the shared MoaB/Mog fold (PF00994). They provide the **MogA-equivalent** adenylyltransferase in the absence of a dedicated MogA. The EC 2.7.7.75 *activity* assignment is a family-level inference (precedent: catalytically competent *Pyrococcus furiosus* MoaB); which paralog is physiologically dominant is unknown → `candidate_uncertain`. - **MocA ×2 (PP_2483, PP_4230):** both K07141 / EC 2.7.7.76 by KEGG orthology, delivered to curation as "MobA-like NTP transferase domain protein." **Important domain caveat:** InterPro shows these carry *only* the generic PF12804/IPR025877 "MobA-like NTP transferase" domain — there is **no MocA-specific InterPro/Pfam signature** in existence, and they lack the MobA-specific IPR013482 that the

true MobA (PP_3457) carries. So the MocA (MCD) upgrade rests on **KEGG orthology (K07141) + genomic adjacency to MCD-molybdoenzyme clusters**, not on a domain signature. Recommendation: re-annotate to MocA as a **moderate-confidence** call (they are clearly *not* MobA and are orthologous to characterised MocA; MocA vs MobA differ by only ~22% identity and are nucleotide-specific — Neumann 2009/2011), and confirm by SSDB reciprocal-best-hit or activity assay.

Direct target-strain evidence. KT2440 functions as a heterologous host that assembles **catalytically fully competent MCD-containing** molybdenum hydroxylases (quinaldine 4-oxidase; quinoline/isoquinoline 2-oxidoreductases), with CMP release on acid hydrolysis confirming the MCD form

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This is strong, *direct* evidence that KT2440's endogenous Moco machinery reaches Mo-MPT **and** can make MCD (MocA active). One caveat from the same work: heterologous Qor in KT2440 pUF1 showed partial Mo-centre deficiency

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i.e. cofactor supply can be limiting for high-level heterologous expression — a physiological, not a genetic, gap.

5. Gaps, ambiguities, and likely over-annotations

- **Not gaps (resolved):** MoeB (PP_0735) and MoaD (PP_1293) — present; only *missing from the candidate list* due to sulfur-relay bucketing.
- **Genuine mechanistic ambiguity:** the MPT-adenylation step. No MogA; the MoaB paralogs carry EC 2.7.7.75 by family inference. Recommend candidate_uncertain.
- **Likely under-annotation (moderate confidence):** PP_2483 and PP_4230 as generic "MobA-like NTP transferase" — should be MocA (EC 2.7.7.76) per KEGG orthology K07141 and MCD-

cluster adjacency. Note: no MocA-specific InterPro domain exists (both share only PF12804 with MobA), so this is an orthology/context call, not a domain call.

- **Domain-shared / disambiguation-by-orthology:** MoeB (PP_0735) sits in the PF00899 ThiF/MoeB/HesA superfamily shared with thiamine-biosynthesis ThiF and other E1-like adenylyltransferases; the specific MoaD-adenylyltransferase assignment relies on KO K21029 (over-propagation watch-point). Domain checks otherwise *confirm* the core calls: MoaD (PP_1293) ThiS/MoaD β -grasp sulfur carrier (PF02597/IPR044672); MoeA (PP_2123) full MoeA architecture (PF00994+PF03453+PF03454); MoaB proteins carry MoaB-specific IPR012245/IPR013484 (i.e. genuinely MoaB, hardening the MogA-absence conclusion).
 - **Likely over-broad bucketing (not over-propagation of function, but pathway assignment):** the ppu00790 seeding pulled in ~21 non-Moco genes (folate, queuosine, riboflavin, bioppterin). These are correctly annotated as their own enzymes but do **not** belong to the Moco module.
 - **GenBank name inconsistency:** KEGG GenBank product strings are unreliable for this locus set (e.g. PP_4597 mislabelled as cPMP synthase; PP_1292 called an "accessory protein"). Curation must use KO/UniProt.
 - **Paralog over-transfer risk:** treating the three MoaA / two MoaB / two MocA copies as functionally identical could over-state redundancy; the island-embedded copies may be client-specialised.
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6. Module and GO-curation recommendations

Module step	Recommended mark	Rationale
MoaA GTP cyclization	<code>covered</code>	PP_4597 + paralogs, SwissProt
MoaC cPMP synthesis	<code>covered</code>	PP_1292, SwissProt
MoeB MoaD activation	<code>covered</code>	PP_0735; add to module (was outside bucket)
MoaD sulfur carrier	<code>covered</code>	PP_1293; add to module (was outside bucket)
MoaD-MoaE synthase	<code>covered</code>	PP_1294 (+PP_1293)
MPT adenylylation (MogA/MoaB)	<code>covered</code> + <code>candidate_uncertain</code>	MoaB lineage substitutes for MogA; competence inferred
MoeA molybdotransfer	<code>covered</code>	PP_2123
MGD (MobA)	<code>covered</code>	PP_3457, SwissProt
MCD (MocA)	<code>covered</code> (branch); gene call moderate-confidence	PP_2483, PP_4230 re-annotate from "MobA-like" → MocA via K07141 + MCD-cluster context (no MocA-specific domain exists)
Eukaryotic fusion variants	<code>not_expected_in_target_taxon</code>	bacterium uses discrete genes

- **Module boundaries:** the *generic module* boundaries are **correct** for KT2440 (both dinucleotide branches apply). No `module_needs_revision` at the biology level. The **seeding query is the problem**: reseed from KEGG module **M00880** (or GO:0006777) rather than the ppu00790 folate map to avoid dragging in folate/queuosine/riboflavin/biopterin genes.
- **Add to module document:** MoeB (PP_0735) and MoaD (PP_1293) as explicit members of the sulfur-carrier activation step.

- **GO requests:** existing terms suffice — GO:0006777 (Mo-molybdopterin biosynthesis), GO:0061599 (molybdopterin adenylyltransferase, for the MogA/MoaB step), GO:0061602 (MoaA GTP 3',8-cyclase), GO:0035673 (MoaA), GO:0061604 (MocA / MCD synthase). No new GO term appears necessary. If a term is needed, a "MoaB-acting-as-MPT-adenylyltransferase" refinement note could help capture the MogA-less lineage.

7. Genes to promote to full fetch-gene review

1. **PP_0735 (moeB)** — sulfur-carrier activation; missing from candidate list; confirm EC 2.7.7.80.
2. **PP_1293 (moaD)** — MPT-synthase small subunit; missing from candidate list.
3. **PP_2483 and PP_4230 (MocA)** — re-annotate from "MoaA-like NTP transferase" to MocA (EC 2.7.7.76); island context supports client-specific MCD synthesis.
4. **PP_2122 and PP_4600 (moaB-I/II)** — resolve which MoaB (if either) is the physiological MPT-adenylyltransferase in the absence of MogA; this is the module's only mechanistic uncertainty.
5. **PP_1969 and PP_2482 (MoaA paralogs)** — clarify whether these are housekeeping-redundant or client-specialised, and whether any is a pseudogene/divergent.

8. Key references

- Parschat et al. 2003, *J Bacteriol* — quinaldine 4-oxidase; KT2440 as host makes catalytically competent **MCD** molybdenum hydroxylase.

P 12730200

- Frerichs-Deeken et al. 2003, *Biochemistry/Eur J Biochem* — quinoline 2-oxidoreductase in KT2440; Mo-MCD form; partial Mo-centre deficiency in KT2440 host.

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- Israel et al. 2002 — isoquinoline 1-oxidoreductase (Mo-MCD) expression in *P. putida*.

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- Bläse et al. 1996 — quinoline 2-oxidoreductase genes; active enzyme in recombinant KT2440.

P 8798497

- Guo et al. 2026 — xanthine oxidase (molybdoenzyme) production in a KT2440 cell factory; Mo-content dependence.

P 41933999

- Neumann et al. 2009, *J Biol Chem* — **MocA** is a specific CTP:molybdopterin cytidyltransferase for **MCD**; distinct from MobA.

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- Neumann et al. 2011 — MobA vs MocA nucleotide-specificity determinants (MGD vs MCD).

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- Data resources: UniProt proteome **UP000000556**; KEGG organism **ppu**, module **M00880**, orthologies K03639/K03637/K03636/K21029/K03635/K03638/K03750/K03752/K07141; InterPro/Pfam domain signatures (PF00994 MoaB/Mog; IPR012245/IPR013484 MoaB-proteobacterial; IPR013482 MobA guanylyltransferase; PF12804/IPR025877 MobA-like NTP transferase; PF00899 ThiF/MoeB; PF02597 ThiS/MoaD; PF03453/PF03454 MoeA).

Uncertainty / species-transfer notes

- Step-to-gene mapping (§3) is **direct** for KT2440 (its own UniProt/KEGG annotations; several SwissProt-reviewed).
- MCD-branch activity is supported by **direct KT2440 experiments** (heterologous MCD-hydroxylase maturation).
- MGD-branch client presence (nitrate reductase, formate dehydrogenases) is inferred from the KT2440 proteome (annotation-level), not from a dedicated MGD-quantification experiment.
- The MoaB-for-MogA adenylation assignment is a **family/homology inference** (mechanistic precedent from *P. furiosus* MoaB and *E. coli* MogA/MoaB biochemistry) — the weakest link and the top experimental question (in-vitro MPT-adenylyltransferase assay of PP_2122 vs PP_4600).

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